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Topic: Fire and Smoke Detection System using Arduino

AIM:

To design and implement a wireless fire and smoke detection system using Arduino and appropriate sensors to detect smoke or fire conditions and raise alerts.

Introduction:

A fire and smoke detection system is an essential safety device used in homes, offices, and industries to prevent fire hazards. Using sensors like **MQ-2** (for smoke detection) and **Flame Sensor** (for fire detection), along with an **Arduino microcontroller**, such systems can identify early signs of fire or smoke and activate alarms or send notifications.

This experiment demonstrates how wireless sensor technology can be used for real-time monitoring of environmental hazards.

Components Used:

Component	Description
Arduino Board (Uno/Nano)	Acts as the central controller to read sensor values and trigger actions.
MQ-2 Gas/Smoke Sensor	Detects smoke, LPG, propane, and other gases.
Flame Sensor	Detects presence of flame using infrared light.
Buzzer	Sounds alarm when smoke or fire is detected.
LED (Optional)	Provides visual alert for fire/smoke detection.
WiFi Module (ESP8266 - Optional)	Sends data to cloud or mobile alert systems.
Power Supply	Provides required voltage to the Arduino and sensors.
Jumper Wires, Breadboard	For circuit connections.

Communication:

The communication between the sensors and the Arduino is handled using **analog and digital signals**:

- **MQ-2 Sensor**: Outputs analog values corresponding to the concentration of gas/smoke.
- **Flame Sensor**: Outputs a digital HIGH/LOW signal depending on flame detection.
- Optional wireless communication using **ESP8266 WiFi module** can send alerts remotely.

Communication Protocols Used:

Protocol	Usage
Analog	Used by the MQ-2 sensor to report varying levels of gas/smoke.
Digital	Used by the flame sensor to indicate flame presence (1 = no flame, 0 = flame detected).
UART (Optional)	If using ESP8266 for cloud communication, UART is used to communicate with Arduino.

Step-by-Step Process:

1. Initialize the Sensors

- Power up the Arduino and initialize serial communication (for debugging).
- Configure the analog pin (MQ-2) and digital pin (Flame Sensor).

2. Read Sensor Data

- Continuously read values from the MQ-2 sensor.
- Check for digital LOW from the flame sensor to detect fire.

3. Process Data

- If the analog value from MQ-2 exceeds a threshold, trigger the buzzer and LED.
- If the flame sensor detects a flame, trigger alerts immediately.

4. Trigger Alerts

- Activate buzzer and LED to indicate danger.
- (Optional) Send message via WiFi module to a cloud platform or smartphone.

5. Reset and Wait

- Once the danger is cleared, system resets and continues monitoring.

Sample Code (Arduino):

```
int smokeA0 = A0;
int flameD0 = 7;
int buzzer = 8;
int led = 9;

void setup() {
  pinMode(flameD0, INPUT);
  pinMode(buzzer, OUTPUT);
  pinMode(led, OUTPUT);
  Serial.begin(9600);
}

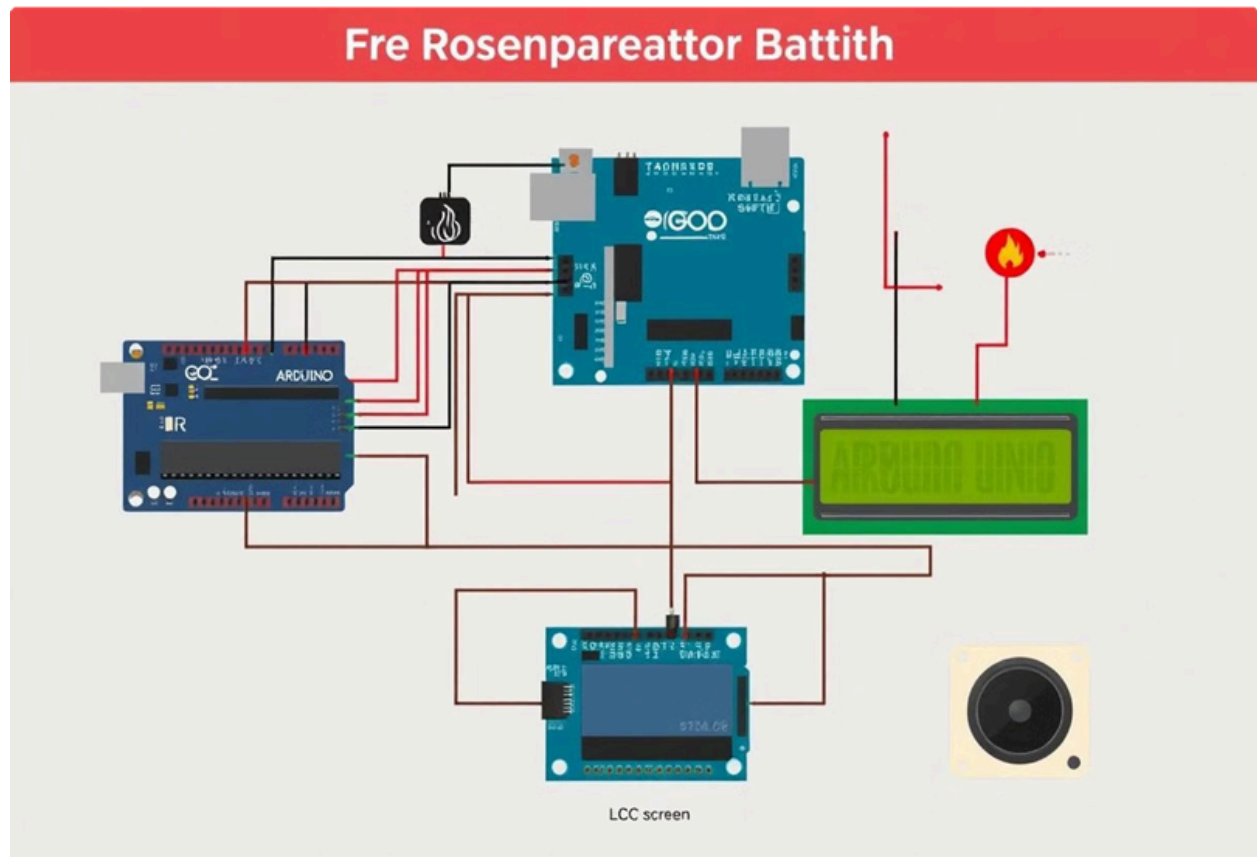
void loop() {
  int smokeLevel = analogRead(smokeA0);
  int flame = digitalRead(flameD0);

  Serial.print("Smoke Level: ");
  Serial.println(smokeLevel);

  if (smokeLevel > 300 || flame == LOW) {
    digitalWrite(buzzer, HIGH);
    digitalWrite(led, HIGH);
    Serial.println("FIRE/SMOKE DETECTED!");
  } else {
    digitalWrite(buzzer, LOW);
    digitalWrite(led, LOW);
  }
}
```

```
delay(1000);  
}
```

Circuit Diagram



Conclusion:

This experiment showcases the integration of gas/smoke and flame sensors with Arduino for building a fire and smoke detection system. The system provides early warnings that can help avoid major hazards. With the addition of wireless modules, the system can be enhanced to send real-time alerts to users via cloud or mobile.